

CMP-12 - Presence and Embodiment Layer

Own bounded Presence, embodiment, audience context, and handoff state while preserving HAL's single identity.

Authority and state ownership

Book I: Decisions 1-7, 24-29, 35, 40, 47-50, and 58. Book II: Chapters 14, 31. Authoritative state: Presence Objects, Embodiment Bindings, Audience Context, Presence Handoffs, modality availability. Explicitly non-owned: HAL identity, Interaction Sessions, credentials, device ownership.

Critical invariant

Every Presence is a bounded manifestation of one HAL identity and cannot independently hold Owner authority.

Logical interfaces

CMP-12-IF-01 / Command / Create Presence

Command `Create Presence` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-12-IF-02 / Command / Bind Embodiment

Command `Bind Embodiment` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-12-IF-03 / Command / Handoff Presence

Command `Handoff Presence` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-12-IF-04 / Command / End Presence

Command `End Presence` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-12-IF-05 / Query / Get Presence Context

Query `Get Presence Context` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-12-IF-06 / Query / List Embodiments

Query `List Embodiments` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-12-IF-07 / Query / Get Audience Context

Query `Get Audience Context` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-12-IF-08 / Event / Presence Created

Event `Presence Created` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-12-IF-09 / Event / Embodiment Bound

Event `Embodiment Bound` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-12-IF-10 / Event / Presence Handed Off

Event `Presence Handed Off` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Proposed > Active > Handoff > Suspended > Ended. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: presence duplication, audience confusion, sensor spoofing, handoff loss, context leakage. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-12-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-12-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-12-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-12-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-12-TST-005 - Critical invariant. Every Presence is a bounded manifestation of one HAL identity and cannot independently hold Owner authority. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-12-TST-006 - Failure containment. Inject presence duplication, audience confusion, sensor spoofing, handoff loss, context leakage. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-12-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-12-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-12-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-12-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.